

BBFlow: Learning Conformational Ensembles of Proteins Based on Backbone Geometry

A Comprehensive Technical Survey

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Abstract

Proteins are dynamic entities whose biological function is governed by the ensemble of conformations they adopt at thermal equilibrium. Molecular Dynamics (MD) simulation provides the rigorous route to sampling this Boltzmann distribution, but at computational costs that often make large-scale application infeasible. Recent deep generative approaches emulate MD by learning from trajectory data, yet the dominant paradigm fine-tunes large pre-trained folding models (AlphaFold 2, ESMFold) and consumes evolutionary sequence information from Multiple Sequence Alignments (MSAs), limiting applicability to natural proteins and incurring prohibitive inference costs. This document provides a comprehensive technical exposition of BBFlow, a conditional $SE(3)$ flow-matching model that generates MD-like protein backbone conformational ensembles from the equilibrium backbone structure alone — without MSAs, language-model embeddings, or pre-trained folding weights. Two key innovations drive the method: (1) a geometric encoding of the equilibrium structure as invariant pairwise distances and equivariant local directions, and (2) a conditional prior distribution on $SE(3)^N$ based on geodesic interpolation toward the equilibrium structure. BBFlow achieves 41× speedup over the best accuracy-comparable baseline (AlphaFlow-T), generalises to de novo proteins and multi-chain complexes without modification, and was trained from scratch in three GPU-days.

1 Introduction

The determination of protein structure has been one of the defining scientific challenges of the past century. The advent of AlphaFold 2 and related methods has largely resolved the static structure prediction problem: given an amino-acid sequence, an accurate equilibrium structure can now be predicted in seconds. However, protein function does not reside in a single static conformation. Enzymes use induced-fit and conformational selection to bind substrates; allosteric proteins communicate regulatory signals via long-range correlated motions; intrinsically disordered regions adopt transient structure upon binding partners. In each case it is the ensemble of accessible conformations — and the free-energy landscape connecting them — that encodes biological activity.

Molecular Dynamics (MD) simulation integrates Newton's equations of motion for all atoms, sampling from the Boltzmann distribution at fixed temperature and pressure. Although MD is physically rigorous, covering the conformational state space exhaustively requires long simulation times (microseconds to seconds for biologically relevant transitions), while each integration step is on the order of femtoseconds. A typical soluble protein of ~300 residues requires hundreds of CPU-hours per 100 ns trajectory. Three 100-ns trajectories per protein — the ATLAS benchmark standard — cost roughly 300 CPU-hours per protein, making large-scale screening computationally prohibitive.

Recent machine-learning MD emulators address this gap by training generative models on pre-computed trajectory data and generating new conformations at inference without simulating physics. The dominant strategy repurposes protein folding architectures: AlphaFlow fine-tunes AlphaFold 2, ConfDiff uses AlphaFold 2's Evoformer representation, and BioEmu builds an AlphaFold-like model trained on heterogeneous trajectory data. While accurate, these approaches are slow (MSA computation or Evoformer processing), potentially biased toward natural proteins for which evolutionary data is abundant, and inapplicable to de novo designed proteins that lack evolutionary history.

1.1 Main Contributions of BBFlow

- Geometry-only conditioning. The equilibrium backbone structure x_{eq} encodes the fold as invariant pairwise $C\alpha$ distances and equivariant local directions — no MSA, no language-model embedding, no pre-trained weights required.
- Conditional prior via geodesic interpolation. Rather than starting ODE integration from unconditioned noise, BBFlow samples x_0 by geodesically interpolating 20% of the way from random noise toward x_{eq} on $SE(3)^N$, providing the correct fold topology as a starting point.
- State-of-the-art speed/accuracy trade-off. 41× faster than AlphaFlow-T at comparable or superior accuracy across five of six standard ensemble metrics on the ATLAS benchmark.
- Generalisation to de novo proteins. No degradation on proteins with no evolutionary history, unlike AlphaFlow (no templates) whose RMSF correlation collapses from 0.86 to 0.47.
- First ensemble model for multi-chain proteins. Omitting residue-index features and conditioning purely on geometry enables zero-shot transfer from monomers to dimers and trimers — a setting no prior model had addressed.
- Trained from scratch. No pre-trained folding weights; 3 GPU-days on 2× NVIDIA A100-40 GB, 18.2 M parameters — orders of magnitude lighter than AlphaFold 2 (93 M parameters, months of training on PDB + UniRef).

2 Background and Prerequisite Concepts

2.1 Protein Structure: Backbone Frames on $SE(3)^N$

A protein is a polypeptide chain of N amino-acid residues drawn from an alphabet of 20 genetically encoded types. Adjacent residues are covalently linked by peptide bonds, forming a backbone of repeating $N-C\alpha-C$ units. The local geometry is described by the dihedral angles ϕ (rotation around $N-C\alpha$) and ψ (rotation around $C\alpha-C$); the ω angle (rotation around $C-N$) is approximately fixed at 180° due to resonance. Allowed (ϕ, ψ) regions define the Ramachandran plot, with α -helices near $(-60^\circ, -45^\circ)$ and β -strands near $(-120^\circ, +120^\circ)$.

In the frame-based representation introduced by AlphaFold 2, each residue i is assigned a local coordinate frame constructed from the $N, C\alpha, C$ atom positions via the Gram–Schmidt process:

$$T_i = (R_i, z_i) \in SE(3), \quad R_i \in SO(3), \quad z_i \in \mathbb{R}^3 \text{ (} C_\alpha \text{ position)} \quad (1)$$

The complete backbone of a protein with N residues is the ordered tuple:

$$x = (T_1, T_2, \dots, T_N) \in SE(3)^N \quad (2)$$

a point on the $6N$ -dimensional Riemannian product manifold $M \equiv SE(3)^N$. The frame representation is coordinate-free (relative geometry is read from $T_i^{-1}T_j$ without a global frame), supports geometrically meaningful interpolation via geodesics, and provides a natural basis for equivariant message-passing.

2.2 Conformational Ensembles and the Boltzmann Distribution

At thermodynamic equilibrium, the probability of observing a protein in conformation x is governed by the Boltzmann distribution:

$$p(x) \propto \exp\left(-\frac{E(x)}{k_B T}\right) \quad (3)$$

where $E(x)$ is the potential energy, k_B is Boltzmann's constant, and T is absolute temperature. The free-energy landscape $F(x) = -k_B T \ln p(x)$ encodes both equilibrium thermodynamics and kinetics: local minima are metastable states, barriers determine transition rates, and the width of a basin determines flexibility. The conformational ensemble is the set of conformations weighted by $p(x)$. Thermodynamic quantities — free energies, entropies, binding affinities — are expectations under this distribution and cannot be obtained from a single static structure alone.

2.3 Molecular Dynamics Simulation

MD numerically integrates Newton's equations for all n atoms:

$$m_k \ddot{r}_k = -\nabla_{r_k} E(r_1, \dots, r_n), \quad \text{for all } k \quad (4)$$

using a molecular force field — an empirically parameterised potential combining bonded terms (harmonic bonds, angles, periodic dihedrals) and non-bonded terms (Lennard-Jones van der Waals, Coulomb electrostatics). The ATLAS dataset used to benchmark BBFlow was generated with GROMACS using the CHARMM36m force field and explicit TIP3P water at 310 K. At a 2-fs integration timestep, 100 ns of simulation requires 5×10^7 force evaluations. For a 300-residue protein in explicit solvent ($\sim 30,000$ atoms) this costs ~ 100 CPU-hours, placing whole-proteome ensemble annotation far beyond current budgets.

Short-timescale MD (100–300 ns) is insufficient for large-scale conformational transitions (folding, allosteric switching) but captures: (i) local backbone and side-chain flexibility; (ii) binding-site plasticity relevant to drug design; (iii) correlated motions underlying allosteric communication; (iv) secondary-structure fraying. These are the properties BBFlow targets.

2.4 The Lie Groups SO(3) and SE(3)

SO(3) is the group of 3×3 rotation matrices with unit determinant. SE(3) is the semidirect product of SO(3) with $\mathbb{R}^3$ — the group of rigid-body motions:

$$SE(3) = SO(3) \ltimes \mathbb{R}^3, \quad (R_1, z_1) \cdot (R_2, z_2) = (R_1 R_2, z_1 + R_1 z_2) \quad (5)$$

A Riemannian metric on $SE(3)^N$ is needed to define geodesics and tangent vectors for flow matching. Following FrameFlow (Yim et al., 2023), the metric is a product: the rotation part uses the bi-invariant trace metric on $so(3)$, and the translation part uses the standard Euclidean metric. Geodesics on $SE(3)^N$ decompose per-residue into independent rotation and translation geodesics. The geodesic from R_0 to R_1 at interpolation fraction t is:

$$R_t = R_0 \cdot \exp(t \log(R_0^T R_1)) \quad (6)$$

where R_0^T denotes the transpose. The tangent vector at R_t pointing toward R_1 is $\text{Log}_{\{R_t\}}(R_1) = R_t \log(R_t^T R_1)$. For translations the geodesic is the straight line $z_t = (1-t)z_0 + t z_1$ with constant tangent $z_1 - z_0$.

2.5 SE(3)-Equivariance in Neural Networks

A function f is G -equivariant if applying the group action g before f equals applying the corresponding transformed output after:

$$f(\rho_{\text{in}}(g) \cdot x) = \rho_{\text{out}}(g) \cdot f(x) \quad \forall g \in G, x \in \mathcal{X} \quad (7)$$

For molecular geometry, $G = SE(3)$. Physical observables must be invariant or covariant under global rotations and translations — there is no privileged coordinate frame in space. Key SE(3)-equivariant architectures include Tensor Field Networks (Thomas et al., 2018), SE(3)-Transformers (Fuchs et al., 2020), E(n) Equivariant GNNs (Satorras et al., 2021), and Frame Averaging (Puny et al., 2022). BBFlow builds on the GAFL architecture, which achieves equivariance via Clifford Frame Attention on SE(3) backbone frames.

2.6 Transformer Self-Attention and Invariant Point Attention

The transformer (Vaswani et al., 2017) processes token embeddings $\{h_i\} \in \mathbb{R}^d$ via scaled dot-product self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (8)$$

where $Q = HW_Q$, $K = HW_K$, $V = HW_V$ are linear projections. The query q_i encodes what residue i seeks; key k_j encodes what residue j provides; value v_j carries the information transferred when attention is high.

Invariant Point Attention (IPA), introduced in AlphaFold 2, extends scalar attention to 3D geometry. Each residue learns local 3D query and key points projected to global coordinates via the residue frame T_i . The attention logits include squared Euclidean distances between global query and key points — invariant to global SE(3) transformations. Output points are rotated back into local frames, making IPA as a whole SE(3)-equivariant. IPA is the core

architectural primitive for FrameDiff, FrameFlow, GAFL, and BBFlow.

2.7 Clifford Frame Attention (CFA)

GAFL (Wagner et al., NeurIPS 2024) extends IPA to Clifford Frame Attention (CFA), representing geometric features — points, lines, planes, and rigid frames — as elements of the projective geometric algebra (PGA) $G(R^{3,0,1})$. In PGA, grade-0 elements are scalars, grade-1 are planes, grade-2 are lines, grade-3 are points, and motors (even subalgebra elements) encode rigid-body motions as sandwich products: $p' = M p M_{\text{rev}}$ (where M_{rev} is the reverse of motor M).

In CFA, residue frames T_i are represented as motors M_i . Edge features between residues i and j are multivectors constructed from the relative motor $M_i^{-1}M_j$. Messages between residues use bilinear products of the algebra (outer, inner, geometric product), enabling higher-order geometric interactions beyond scalar inner products. This is the architecture inherited by BBFlow’s flow vector-field network.

2.8 Flow Matching on Riemannian Manifolds

Flow matching (Lipman et al., ICLR 2023; Chen & Lipman, ICLR 2024) learns a time-dependent vector field $v_\theta : M \times [0, 1] \rightarrow TM$ that transports samples from a prior p_0 to the data distribution p_1 via an ODE:

$$\frac{dx_t}{dt} = v_\theta(x_t, t), \quad x_0 \sim p_0 \quad (9)$$

Training is made tractable by conditional flow matching (CFM): connect individual data pairs (x_0, x_1) by a simple conditional path $\psi(x_0, x_1, t)$, compute the conditional vector field u_t analytically, and regress:

$$\mathcal{L}_{\text{FM}} = \mathbb{E} [\|v_\theta(x_t, t) - u_t\|^2], \quad x_t = \psi(x_0, x_1, t) \quad (10)$$

The conditional and marginal losses share the same gradient, so minimising the conditional loss yields the correct marginal vector field. On manifolds M , straight paths are replaced by geodesics $\gamma(x_0, x_1, t)$ and u_t is the geodesic velocity. On $SE(3)^N$ the ground-truth conditional vector fields are:

$$v_{SO(3)}(R_t, t \mid R_1) = \frac{\text{Log}_{R_t}(R_1)}{1-t} \quad (11)$$

$$v_{\mathbb{R}^3}(z_t, t \mid z_1) = \frac{z_1 - z_t}{1-t} \quad (12)$$

Key advantages over diffusion: (i) straight geodesic paths allow accurate ODE integration in as few as 20 steps (vs. 50–1000 for diffusion); (ii) no Gaussian prior requirement — any p_0 can be used, enabling BBFlow’s conditional prior; (iii) the Riemannian extension is analytically cleaner than heat-kernel diffusion.

3 Prior Work: Approaches to Conformational Ensemble Generation

3.1 Classical and Semi-Classical Methods

Molecular Dynamics (Gold Standard)

MD is physically rigorous and requires no training data. Its primary limitation is cost: $O(10^4\text{--}10^6)$ CPU-hours per protein for trajectories long enough to capture biologically relevant transitions. Even short-timescale MD (100–300 ns) costs ~100–300 CPU-hours per protein, making proteome-scale screening impractical.

Normal Mode Analysis (NMA, Case 1994)

NMA approximates the energy surface near equilibrium as a quadratic and analytically computes normal modes — directions of concerted motion with definite frequencies. NMA is extremely fast (minutes per protein) but is a linear approximation that systematically underestimates ensemble spread and cannot capture anharmonic or large-amplitude motions. Benchmarks show NMA consistently worse than all generative baselines on ATLAS metrics.

MSA Subsampling (Wayment-Steele et al., Nature 2024)

By running AlphaFold 2 with different MSA subsets, one can coax alternative conformations. This is powerful for proteins whose conformational substates are encoded in evolutionary variation, but is limited to proteins with rich MSA coverage and cannot sample thermally accessible states not reflected in sequence variation. It also inherits the full cost of MSA construction and AlphaFold 2 inference.

3.2 Boltzmann Generators (Noé et al., Science 2019)

Boltzmann Generators (BGs) train normalising flows to map a Gaussian latent space directly to the Boltzmann distribution $p(x) \propto \exp(-E(x)/k_{\text{BT}})$, combining maximum likelihood on MD samples with physics-based free-energy minimisation. BGs can in principle sample thermodynamically equilibrated configurations including rare states not visited by short MD.

Critical limitation: BGs must be trained per protein. The normalising flow is fitted to the specific energy landscape of one molecule and is not transferable. A model trained on ubiquitin cannot generate ensembles for lysozyme without retraining. This severely limits practical utility for screening — each new target requires a fresh training run with sufficient MD data.

3.3 AlphaFlow: AlphaFold-Based Ensemble Generation (Jing et al., ICML 2024)

AlphaFlow proposed the first transferable protein ensemble generation model by reformulating ensemble generation as a structure denoising task within AlphaFold 2. Four variants are published:

- AlphaFlow (no templates). Pre-trained AlphaFold 2 is fine-tuned on ATLAS trajectory data. At inference the model predicts a conformation from sequence + MSA, starting from a noisy structure. MSA computation is required per sequence. Time: 32.0 s/conf.
- AlphaFlow-T (with templates). The equilibrium structure x_{eq} is provided as a template to AlphaFold's template processing module, substantially improving accuracy. A full AlphaFold forward pass is required at each of the 10 flow timesteps. Time: 32.6 s/conf.
- ESMFlow-T. Uses ESMFold protein language model (PLM) embeddings instead of the MSA-based Evoformer, avoiding per-sequence MSA computation but still depending on PLM weights

trained on natural proteins. Time: 11.2 s/conf.

- Distilled variants (AlphaFlow-T_dist, AlphaFlow-T_12L_dist). Consistency distillation reduces timesteps from 10 to 1; the 12-layer variant also shrinks depth. Times: 3.3 s and 1.2 s/conf respectively, at reduced accuracy.

Core limitations: (1) evolutionary-information dependency — even AlphaFlow-T uses MSA or PLM embeddings unavailable for de novo proteins; (2) inference cost — 32.6 s/conf; (3) over-stabilisation — AlphaFlow-T predicts median RMSF 1.17 Å vs. MD reference 1.48 Å, systematically under-exploring conformational space; (4) single-chain only.

3.4 ConfDiff: Force-Guided SE(3) Diffusion (Wang et al., ICML 2024)

ConfDiff replaces AlphaFold 2's structure module with an SE(3) diffusion model conditioned on the pre-trained Evoformer embedding of the input sequence, optionally augmented with a force-field energy guidance signal. The energy guidance improves physical plausibility of generated conformations. Like AlphaFlow, ConfDiff requires the Evoformer (hence MSA), has inference time 20.2 s/conf, and underperforms AlphaFlow-T on ATLAS metrics in most categories.

3.5 BioEmu: Scalable Equilibrium Ensemble Emulation (Lewis et al., Science 2025)

BioEmu trains an AlphaFold-like model on a large heterogeneous dataset of MDs of varying lengths and temperatures plus NMR and crystallographic ensemble data. It targets general equilibrium ensembles — including rare alternative folded states — rather than the specific distribution induced by a fixed MD protocol. Because it is not trained to reproduce the ATLAS distribution, it performs unfavourably on the ATLAS benchmark (RMSF MAE 1.29 Å vs. BBFlow's 0.42 Å), though this is expected given the mismatch in training objectives. BioEmu still relies on MSA and an AlphaFold-like architecture with inference time 1.9 s/conf.

3.6 MDGen: Time-Consistent All-Atom Trajectories (Jing et al., NeurIPS 2024)

MDGen models the joint distribution over consecutive MD frames rather than independent snapshots, generating time-consistent all-atom trajectories in a token-based discrete architecture. Although fast (0.15 s/conf) and trained on ATLAS, MDGen achieves substantially lower ensemble accuracy than BBFlow: RMSF MAE 0.81 vs. 0.42 Å; DCCM correlation 0.54 vs. 0.87.

3.7 The Gap BBFlow Addresses

Surveying prior work reveals four simultaneous limitations that no existing method solves:

- Accurate but slow: AlphaFlow-T reaches RMSF $r = 0.92$ but requires 32.6 s/conf and MSA.
- Fast but inaccurate: AlphaFlow-T_12L_dist (1.2 s) and BioEmu (1.9 s) sacrifice quality.
- No multi-chain support: residue-index features in all prior models prevent application to protein complexes.
- Evolutionary-information dependency: all accurate models fail for de novo proteins lacking MSA coverage.

BBFlow addresses all four simultaneously by conditioning on backbone geometry rather than sequence, introducing a conditional prior that reduces the flow's learning burden, and omitting residue-index features to enable geometry-only chain-agnostic inference.

4 BFlow: Method

4.1 Problem Formulation

BFlow models the conformational ensemble as the conditional distribution $p(x | x_{\text{eq}})$, where $x \in \text{SE}(3)^N$ is a backbone conformation and $x_{\text{eq}} \in \text{SE}(3)^N$ is the equilibrium backbone structure. This formulation is natural for MD emulation since MD itself also requires an initial (equilibrium) structure. Conditioning on x_{eq} rather than the sequence encodes the fold identity without requiring evolutionary data.

$$x \sim p(x | x_{\text{eq}}) \approx \phi_1(x_0; x_{\text{eq}}), \quad x_0 \sim p_0(\cdot | x_{\text{eq}}) \quad (13)$$

where ϕ_t is the flow endpoint ($t = 1$ solution of the ODE starting from x_0) and $p_0(\cdot | x_{\text{eq}})$ is the novel conditional prior described in Section 4.4.

4.2 Conditional Flow Matching for Ensemble Generation

Extending the flow ODE to include the conditioning signal x_{eq} , BFlow learns:

$$v(x, t, x_{\text{eq}}) : \mathcal{M} \times [0, 1] \times \mathcal{M}_{\text{eq}} \rightarrow T_x \mathcal{M} \quad (14)$$

defining the conditional flow $\phi_t(\cdot | x_{\text{eq}})$ via:

$$\frac{d}{dt} \phi_t(x | x_{\text{eq}}) = v(\phi_t, t, x_{\text{eq}}), \quad \phi_0(x | x_{\text{eq}}) = x \quad (15)$$

The conditional training loss is:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}[\|v - \hat{v}(x_t, t, x_{\text{eq}})\|_{\text{SE}(3)}^2] \quad (16)$$

The expectation is over $t \sim \text{U}(0, 1)$, $(x_1, x_{\text{eq}}) \sim p_{\text{data}}$, $x_0 \sim p_0(\cdot | x_{\text{eq}})$, $x_t = \gamma(x_0, x_1, t)$ (geodesic interpolation), and v is the ground-truth conditional vector field from Equations (11)–(12). The $\text{SE}(3)$ norm decomposes as:

$$\|v\|_{\text{SE}(3)}^2 = \frac{1}{2} \text{Tr}(v_r v_r^T) + \|v_z\|^2 \quad (17)$$

measuring rotational and translational components independently. The network predicts $\hat{x}_1 = \hat{x}_\theta(x_t, t, x_{\text{eq}})$ and the predicted field $\hat{v}$ is recovered from $\hat{x}_1$ via Equations (11)–(12).

4.3 Encoding the Equilibrium Structure

The equilibrium structure $x_{\text{eq}} = \{(R_i^{\text{eq}}, z_i^{\text{eq}})\}$ is encoded as initial edge features of the message-passing network via two complementary representations.

Distance encoding

Inspired by the interpretation of evolutionary co-variation as a learned contact map, pairwise Ca – Ca distances are binned into 22 uniform bins spanning $[0, 20]$ Å:

$$s_{ij} = \text{bin}(\|z_i^{\text{eq}} - z_j^{\text{eq}}\|_2) \in \mathbb{R}^{22} \quad (18)$$

This 22-dimensional one-hot edge feature is invariant to global rigid-body transformations and provides coarse fold topology information analogous to a contact map.

Direction encoding

For residue pairs within 5 Å, the unit vector from residue i to residue j is expressed in the local co-rotating frame of residue i :

$$e_{ij} = (R_i^{\text{eq}})^{-1} \cdot \frac{z_i^{\text{eq}} - z_j^{\text{eq}}}{\|z_i^{\text{eq}} - z_j^{\text{eq}}\|_2} \in \mathbb{R}^3 \quad (19)$$

Transforming into the local frame makes the direction components invariant to global rotations. Together with s_{ij} , this provides geometrically richer information analogous to equivariant tensor-network features, enabling better discrimination of local structural context.

Amino-acid identity

A 20-dimensional one-hot encoding of residue type is projected to a 128-dimensional node embedding by a linear layer. This encodes local backbone degrees of freedom (proline is rigid, glycine is highly flexible) without providing global sequence context. An ablation (Section 7) shows the model remains competitive even without amino-acid identity, demonstrating that backbone geometry alone suffices.

4.4 Conditional Prior Distribution

Standard flow matching uses an unconditional prior: Gaussian translations and uniform rotations. The flow must then assemble a coherent fold from complete noise — a challenging task that increases variance. BBFlow introduces a conditional prior $p_0(\cdot | x_{\text{eq}})$ based on geodesic interpolation on $\text{SE}(3)^N$:

$$X_{\text{uncond}} \sim p_{\text{uncond}}, \quad X_0 = \gamma(X_{\text{uncond}}, X_{\text{eq}}, \xi) \quad (20)$$

where γ is the product geodesic on $\text{SE}(3)^N$ (factoring per-residue into rotation and translation geodesics) and $\xi \in (0, 1)$ is a hyperparameter:

$$\gamma(X_{\text{uncond}}, X_{\text{eq}}, 0) = X_{\text{uncond}}, \quad \gamma(X_{\text{uncond}}, X_{\text{eq}}, 1) = X_{\text{eq}} \quad (21)$$

At $\xi = 0.2$ (used in all main experiments), the prior sample is 20% of the geodesic distance toward the equilibrium structure — the backbone topology is approximately correct, but with enough noise diversity to generate varied conformations. This is a principled generalisation of partial denoising from diffusion models to the manifold flow matching framework. An ablation study of ξ (Section 7.2) confirms that $\xi = 0.2$ optimally balances accuracy and ensemble diversity.

4.5 Network Architecture

The vector field network $\hat{v}_\theta(x_t, t, x_{\text{eq}})$ is adapted from GAFL (Wagner et al., NeurIPS 2024), itself an extension of FrameDiff (Yim et al., ICML 2023) and FrameFlow (Yim et al., 2023). Six CFA message-passing blocks iteratively update node features, edge features, and backbone frames. Each block: (1) computes geometric messages between all residue pairs using Clifford Frame Attention; (2) aggregates messages to update node features; (3) predicts an incremental frame update; (4) updates edge features from new node features and inter-frame geometry.

Inputs: current frames $\{(R_i^t, z_i^t)\}$, flow time t , amino-acid identity embeddings, and — as initial edge features — equilibrium distance s_{ij}^{eq} (22 dims), equilibrium direction e_{ij} (3 dims), and current pairwise distance s_{ij}^t (22 dims). This expands the edge feature dimension by 25 relative to the unconditional GAFL baseline.

Critical design choice — omitting residue indices: Unlike AlphaFold 2, FrameDiff, FrameFlow, and AlphaFlow, BBFlow does not use absolute residue indices as node features. This (1) reduces the risk of per-residue memorisation from training data; (2) makes the architecture agnostic to chain identity, enabling zero-shot transfer from monomers to multi-chain complexes. The

ablation confirms that adding residue indices does not improve accuracy on monomers but causes failure on multi-chain proteins.

BBFlow has ~18.2 M learnable parameters and was trained for 3 days on two NVIDIA A100-40 GB GPUs, starting from random initialisation (no pre-trained folding weights). BBFlow-light (3 CFA blocks, feature dims 96/48) has 2.5 M parameters and achieves ~200× speedup over AlphaFlow at modestly reduced accuracy.

4.6 Training Algorithm

At each training step: (1) sample (x_{eq}, x_1) from ATLAS; (2) sample $x_0 \sim p_0(\cdot | x_{\text{eq}})$ via geodesic interpolation with $\xi = 0.2$; (3) sample $t \sim U(0, 1)$; (4) compute $x_t = \gamma(x_0, x_1, t)$; (5) compute ground-truth vector field v from Eqs. (11)–(12); (6) compute $\hat{x}_1 = \hat{x}_\theta(x_t, t, x_{\text{eq}})$ and predicted field $\hat{v}$; (7) update θ to minimise $\mathcal{L} = \|v - \hat{v}\|_{\text{SE}(3)}^2 + \mathcal{L}_{\text{aux}}(x_1, \hat{x}_1)$, where $\mathcal{L}_{\text{aux}}$ (from FrameFlow) regresses directly on x_1 and accelerates convergence.

4.7 Inference

Given x_{eq} : (1) sample $x_{\text{uncond}} \sim p_{\text{uncond}}$; (2) compute $x_0 = \gamma(x_{\text{uncond}}, x_{\text{eq}}, 0.2)$; (3) integrate the conditional flow ODE using 20 Euler steps on $\text{SE}(3)^N$ (per-residue rotation update via exponential map + translation update); (4) endpoint $x_1 = \phi_1(x_0 | x_{\text{eq}})$ is one independent conformation sample from $p(x | x_{\text{eq}})$. Repeat steps 1–4 for each desired conformation (all samples are independent given x_{eq}). Using only 20 steps — vs. 50–1000 for diffusion — is possible because flow-matching trajectories are nearly straight geodesics.

5 Experimental Setup

5.1 ATLAS Dataset

ATLAS (Vander Meersche et al., Nucleic Acids Research 2024) provides standardised all-atom MD trajectories for 1390 structurally diverse monomeric proteins from the Protein Data Bank. Each protein has three independent 100-ns simulations at 310 K, CHARMM36m force field, explicit TIP3P water, GROMACS engine. Standardised conditions make ensemble metrics comparable across proteins and models — a key requirement for benchmarking. The same training/validation/test split as AlphaFlow is used: 1265 / 39 / 82 proteins. 250 conformations are generated per protein for evaluation.

5.2 De Novo Protein Dataset

To evaluate generalisation beyond natural proteins, 50 de novo proteins were generated with RFdiffusion and FrameFlow, each subjected to three 100-ns MD simulations following the ATLAS protocol. Equilibrium structures were obtained with ESMFold (the only folding model without an MSA requirement). This dataset is challenging because all MSA-dependent baselines face a near-zero evolutionary signal.

5.3 Multi-Chain Systems

Five well-studied complexes were simulated: barnase–barstar dimer (1BGS), homotrimeric foldon (1RFO), homodimeric fructokinase (5EY7), nanobody–SARS-CoV-2 RBD heterodimer (7KGK), and nanobody–TNFRSF17 heterodimer (8HXR). Total residues range from 80 to 590. BBFlow was applied without any modification despite being trained only on monomers.

5.4 Evaluation Metrics

All metrics use C α atoms after superimposition to the equilibrium structure. MD bootstrapping (100 replicates) estimates uncertainty in MD-derived statistics.

- RMSF (Root Mean Square Fluctuation): per-residue C α deviation magnitude. Reported: Pearson correlation r (profile shape), MAE (amplitude accuracy), and median RMSF (over/under-stabilisation; MD reference = 1.48 Å on ATLAS).
- Pairwise RMSD (pwRMSD): mean C α RMSD across all conformation pairs, quantifying ensemble spread without a reference state. Reported: MAE vs. MD.
- DCCM (Dynamic Cross-Correlation Matrix): DCCM_{ij} = normalised covariance of C α displacements for residues i, j . Values +1 / -1 / 0 indicate correlated / anti-correlated / uncorrelated motion. Reported: Pearson r of flattened matrix.
- PCA Wasserstein-2 (W_2): generated and MD conformations projected onto the first two MD principal components; W_2 distance between the resulting 2D distributions.
- Transient contact Jaccard (J_{tr}): Jaccard similarity between transient contact sets (pairs separated > 8 Å in x_{eq} but < 8 Å in $\geq 10\%$ of ensemble conformations) for MD vs. generated ensembles.

6 Results

6.1 ATLAS Benchmark

Table 1 — ATLAS test set (82 proteins). Median over proteins. RMSF Median: MD reference = 1.48 Å. Inference time: protein 7c45A (302 residues), NVIDIA A100-80 GB. (↑) higher is better, (↓) lower is better.

Method	RMSF r (↑)	RMSF MAE(↓)	PwRMSD(↓)	DCCM r (↑)	PCA W_2 (↓)	$J_{tr}\%$ (↑)	Time(s)(↓)
BioEmu*	0.83	1.29	2.84	0.80	1.65	36	1.9
AlphaFlow	0.86	0.59	1.35	0.86	1.47	41	32.0
ConfDiff	0.88	0.62	1.45	0.86	1.41	39	20.2
AlphaFlow-T	0.92	0.41	0.91	0.89	1.28	47	32.6
ESMFlow-T	0.92	0.52	1.22	0.89	1.48	47	11.2
AlphaFlow-Tdist	0.92	0.68	1.41	0.88	1.43	42	3.3
AlphaFlow-T12L,d	0.90	0.85	1.80	0.87	1.60	24	1.2
BBFlow	0.90	0.42	0.77	0.87	1.33	29	0.8

* BioEmu not trained on ATLAS; comparison is illustrative only. BBFlow row in bold. Median RMSF of BBFlow = 1.49 Å $\approx$ MD reference 1.48 Å.

Key observations from Table 1:

- Speed/accuracy Pareto. BBFlow is 41× faster than AlphaFlow-T (0.8 vs. 32.6 s/conf) while matching or exceeding it on 5 of 6 metrics — the only model below 4 s/conf that achieves competitive accuracy.
- Over-stabilisation in AlphaFlow-T. AlphaFlow-T predicts median RMSF 1.17 Å vs. MD reference 1.48 Å — a 21% underestimate of ensemble spread. BBFlow predicts 1.49 Å, closely matching MD. Over-stabilisation is attributed to the folding model's strong prior toward equilibrium structures.
- Pairwise RMSD. BBFlow achieves the best pwRMSD MAE (0.77 Å), indicating superior capture of the magnitude of conformational changes vs. all baselines including AlphaFlow-T (0.91

Å).

- Transient contacts. BBFlow underperforms MSA-based models on J_{tr} (29% vs. 47%). Transient contacts are rare, large-amplitude excursions best predicted using evolutionary co-variation signals unavailable to BBFlow.

6.2 De Novo Proteins

Table 2 — De novo proteins (50 systems). MD reference RMSF median = 0.91 Å.

Method	RMSF r(↑)	RMSF MAE(↓)	PwRMSD(↓)	DCCM r(↑)	PCA W2(↓)	Jtr%(↑)	Time(s)(↓)
BioEmu*	0.60	4.24	8.29	0.64	1.53	23	1.9
AlphaFlow	0.47	4.76	7.40	0.58	1.64	17	32.0
ConfDiff	0.62	3.82	7.26	0.65	1.72	15	20.2
AlphaFlow-T	0.89	0.25	0.38	0.85	0.66	55	32.6
ESMFlow-T	0.89	0.28	0.43	0.86	0.63	55	11.2
AlphaFlow-Tdist	0.88	0.46	0.77	0.84	0.69	51	3.3
AlphaFlow-T12L,d	0.87	0.58	0.97	0.83	0.75	38	1.2
BBFlow	0.84	0.26	0.32	0.83	0.67	32	0.8

AlphaFlow (no templates) and BioEmu degrade catastrophically without MSA. BBFlow RMSF correlation drops only from 0.90 to 0.84 ($\Delta = 0.06$); AlphaFlow drops from 0.86 to 0.47 ($\Delta = 0.39$). BBFlow achieves the best pairwise RMSD MAE (0.32 Å) among all models.

6.3 Multi-Chain Proteins

Table 3 — Five multi-chain systems. BBFlow trained on monomers only. MD reference RMSF median = 1.20 Å. No other baseline produced physically valid states.

Method	RMSF r(↑)	RMSF MAE(↓)	PwRMSD(↓)	DCCM r(↑)	PCA W2(↓)
BBFlow	0.82	0.31	0.41	0.85	0.71

BBFlow correctly captures both intra-chain fluctuations and inter-chain allosteric communication (DCCM r = 0.85) despite never training on multimers. AlphaFlow with parser-modified multi-chain input generates unphysical states.

6.4 Sequence-to-Ensemble Pipeline

When only a sequence is available, BBFlow can be combined with AlphaFold 2 in a two-stage pipeline: AF2 predicts x_{eq} once, then BBFlow generates all conformations. Because AF2 runs only once (not once per conformation), this pipeline is ~30× faster than AlphaFlow.

Table 4 — Sequence-to-ensemble pipeline, ATLAS test set.

Method	RMSF r	RMSF MAE	PwRMSD	DCCM r	PCA W2	Time(s)
AlphaFlow	0.86	0.59	1.35	0.86	1.47	32.0
AF2 + BBFlow	0.87	0.52	1.07	0.85	1.47	1.1

AF2 + BBFlow outperforms stand-alone AlphaFlow on most metrics and is 29× faster (1.1 s = 0.3 s AF2 + 0.8 s BBFlow vs. 32.0 s AlphaFlow).

7 Ablation Study

7.1 Component Ablations

Table 5 — Component ablations on ATLAS test set. Each variant removes or replaces one

component of BBFlow. Cond.P = conditional prior; Dist = distance encoding; Dir = direction encoding; AA = amino-acid identity; Idx = residue index.

Variant	Cond.P	Dist	Dir	AA	Idx	RMSF MAE(↓)	PwRMSD(↓)
BBFlow	YES	YES	YES	YES	NO	0.42	0.77
(a) no dir.	NO	YES	NO	YES	YES	0.52	1.15
(b) no prior	NO	YES	YES	YES	YES	0.48	0.90
(c) + index	YES	YES	YES	YES	YES	0.42	0.82
(d) no AA	YES	YES	NO	NO	YES	0.54	0.93
(e) no dist.	YES	NO	NO	YES	YES	5.88	7.08

Removing the distance encoding (e) is catastrophic — RMSF MAE 0.42 → 5.88 Å — confirming it as the essential fold-identity signal. Direction encoding (a) and conditional prior (b) each give measurable improvements. Residue indices (c) are neutral for monomers but harmful for multi-chain transfer.

Detailed findings:

- Distance encoding (e) — **ESSENTIAL**. Removing it is catastrophic: RMSF MAE 0.42 → 5.88 Å, PwRMSD 0.77 → 7.08 Å. Without distance encoding the model has no information about which fold to generate; the result is worse than random generation.
- Direction encoding (a) — **IMPORTANT**. Removes fine-grained local geometry: RMSF MAE 0.42 → 0.52 Å, PwRMSD 0.77 → 1.15 Å. The direction vectors encode which way neighbouring residues point within the local frame, enabling better discrimination of structurally similar but dynamically different local environments.
- Conditional prior (b) — **BENEFICIAL**. RMSF MAE 0.42 → 0.48 Å, PwRMSD 0.77 → 0.90 Å. The improvement in final accuracy is modest, but the conditional prior also substantially accelerates convergence during training.
- Residue index (c) — **NEUTRAL** for monomers, **HARMFUL** for multimers. Adding indices does not improve accuracy on the ATLAS benchmark (RMSF MAE identical; PwRMSD slightly worse at 0.82 vs. 0.77 Å). Residue indices cause catastrophic failure on multi-chain proteins because they encode a single-chain assumption.
- Amino-acid identity (d) — **HELPFUL** but not essential. Removing AA identity degrades RMSF MAE 0.42 → 0.54 Å. The model without any sequence information still outperforms AlphaFlow (0.59 Å) and all distilled models, confirming that backbone geometry alone is sufficient for competitive ensemble generation.

7.2 Hyperparameter ξ — Conditional Prior Strength

The hyperparameter ξ controls how close the prior sample x_0 is to the equilibrium structure x_{eq} . Smaller ξ increases ensemble diversity but demands the flow learn a larger transformation from noise; larger ξ constrains diversity. Table 6 shows the inference-time ablation for the model trained at $\xi = 0.2$.

Table 6 — Inference-time ablation of ξ (model trained with $\xi = 0.2$).

ξ	RMSF r	RMSF MAE	RMSF Med.	PwRMSD	DCCM r	PCA W2	Note
0.01	0.70	7.61	11.16	10.83	0.73	2.85	
0.05	0.81	3.89	5.43	5.38	0.78	2.06	
0.10	0.88	1.32	2.62	2.21	0.84	1.51	
0.20	0.90	0.42	1.49	0.77	0.87	1.33	← optimal
0.30	0.89	0.47	1.37	1.02	0.86	1.69	
0.40	0.86	0.64	1.65	1.31	0.80	2.12	
0.50	0.79	0.80	1.82	1.50	0.74	3.00	
0.60	0.73	0.82	1.70	1.65	0.70	4.51	

$\xi = 0.20$ is optimal. Below 0.2 the flow must learn too large a transformation; above 0.2 the prior is too constrained toward x_{eq} , reducing ensemble diversity.

8 Discussion

8.1 What Makes BBFlow Conceptually Novel?

Decoupling ensemble generation from sequence modelling

Every prior transferable ensemble generation model — AlphaFlow, ESMFlow, ConfDiff, BioEmu — inherits the architecture of protein folding models, framing ensemble generation as a sequence-to-structure problem. This coupling is not necessary: the fold of a protein is already fully captured in its equilibrium structure x_{eq} , which can serve as the conditioning signal directly. BBFlow demonstrates that the entire sequence-level machinery (MSA computation, Evoformer, language-model pretraining) is not a prerequisite for high-quality MD emulation. The resulting model is faster, lighter, unbiased toward natural-protein sequence statistics, and applicable to any protein for which a structural model exists.

Conditional prior in Riemannian flow matching

The concept of a geometry-conditioned prior in flow matching is novel. Prior work used either unconditional priors (GAFL, FrameFlow) or non-Gaussian but still unconditional priors (Chroma). BBFlow's conditional prior generalises partial denoising from diffusion models to the manifold flow matching framework, providing a principled geometrically correct way to bias generation toward a known reference structure while retaining sufficient diversity. This concept is broadly applicable to any generative modelling problem on curved spaces where a reference point is available.

Zero-shot multi-chain transfer via geometry-only conditioning

Omitting residue-index features — and conditioning on geometry alone — enables zero-shot generalisation from monomers to multimers. It reveals that the chain-identity information encoded by residue indices is precisely what prevents prior models from handling multi-chain inputs: they bake in a single-chain assumption. By replacing this with the geometry of x_{eq} (which naturally encodes chain boundaries as spatial discontinuities in the distance matrix), BBFlow extends to protein complexes without any architectural modification — the first ensemble model to do so.

8.2 Qualitative Comparison with Baselines

Table 7 — Qualitative comparison of ensemble generation approaches.

Method	MSA	PLM	FT	Struct	Multi	Time(s)	RMSF MAE	Notes
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AlphaFlow	Y	N	Y	N	N	32.0	0.59	Evoformer+MSA
AlphaFlow-T	Y	N	Y	Y	N	32.6	0.41	Over-stabilises
ESMFlow-T	N	Y	Y	Y	N	11.2	0.52	ESM embeddings
ConfDiff	Y	N	Y	N	N	20.2	0.62	Force-guided
BioEmu	Y	N	Y	N	N	1.9	1.29	Non-ATLAS target
MDGen	N	N	N	N	N	0.15	0.81	Time-consistent
BBFlow	N	N	N	Y	Y	0.8	0.42	This work

MSA = requires MSA. PLM = protein language model. FT = fine-tunes pre-trained weights. Struct = conditions on equilibrium structure. Multi = multi-chain support.

8.3 Limitations

- MD distribution emulation. BBFlow reproduces the distribution of 300-ns CHARMM36m GROMACS trajectories at 310 K. It cannot predict rare alternative folded states, millisecond-timescale transitions, protein unfolding, or ligand-binding-induced large-scale conformational changes.
- Transient contact accuracy. Systematic underperformance ($J_{tr} = 29\%$ vs. 47% for AlphaFlow-T) on rare large-amplitude excursions that create new inter-residue contacts. Evolutionary co-variation, which encodes information about alternative structural states, appears especially beneficial for predicting these rare events.
- Backbone only. BBFlow generates C α -only backbone conformations. Side-chain ensembles and protein-ligand interactions require additional modelling.
- Requires an equilibrium structure. This is not a practical limitation for MD emulation (MD also requires an initial structure), but it does require a structure prediction step for purely sequence-based inputs. The AF2 + BBFlow pipeline handles this and remains $29\times$ faster than AlphaFlow.

8.4 Practical Applications

- Drug discovery. Rapid screening of protein flexibility and cryptic-pocket availability across candidate targets at 0.8 s/conformation — feasible for whole-proteome annotation.
- Antibody and nanobody engineering. Multi-chain support enables ensemble generation for antibody-antigen complexes and evaluation of binding-site dynamics.
- De novo protein design. Integration into design pipelines (RFDiffusion + ProteinMPNN + BBFlow) to screen designed sequences for desired dynamic properties — a capability absent from current design workflows.
- Large-scale structural annotation. At 0.8 s/conf, generating 250 conformations for all ~200,000 proteins in the AlphaFold Database requires ~10 GPU-days.

9 Conclusion

This document has provided a comprehensive technical exposition of BBFlow, covering all prerequisite concepts (protein backbone geometry, Boltzmann ensembles, molecular dynamics, SE(3) Lie groups, equivariant neural networks, transformer attention, Clifford Frame Attention, and Riemannian flow matching), a thorough survey of prior ensemble generation approaches (Boltzmann Generators, AlphaFlow/ESMFlow, ConfDiff, BioEmu, MDGen), and a full technical derivation of every BBFlow component.

BBFlow makes three fundamental contributions: (1) it demonstrates that the sequence machinery of protein folding models is unnecessary for high-quality MD emulation — backbone geometry

alone is sufficient; (2) it introduces a conditional prior via geodesic interpolation on $SE(3)^N$, a principled generalisation of partial denoising to manifold flow matching; and (3) it shows that geometry-only conditioning enables zero-shot transfer from monomers to multi-chain complexes.

On the ATLAS benchmark, BBFlow achieves $41\times$ speedup over AlphaFlow-T at comparable or superior accuracy across five of six metrics, correctly calibrates ensemble spread (median RMSF 1.49 Å vs. MD reference 1.48 Å), and is the only model with competitive accuracy that does not degrade on de novo proteins. The model was trained from scratch in three GPU-days with 18.2 M parameters — a fraction of the resources required by AlphaFold-based baselines.

The two core ideas — structure conditioning as a replacement for evolutionary information, and conditional priors in Riemannian flow matching — are broadly applicable beyond protein ensemble generation and may inform future work in structural biology, generative chemistry, and geometric deep learning.

Code and pre-trained weights: <https://github.com/graeter-group/bbflow>

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